Software Requirements Specification (SRS)

Client: University of Pretoria - Biophysics Group

Team Member	Student Number
Kudakwashe Mujuru	u20620111
Carlos Juma	u22602047
Nomzi Phosa	u23577160
Mbali Thobejane	u23543672

Team: CoreTech
Contact: coretech.capstone@gmail.com
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the web-based version of Full_SMS (Single-Molecule Spectroscopy Analysis Software). The document is intended for the development team, project stakeholders, and the client to establish a shared understanding of the system's functionality, constraints, and expected behavior.

1.2 Project Scope

Full_SMS Web is a browser-based scientific data analysis platform designed to improve accessibility to single-molecule spectroscopy (SMS) analysis tools. The system enables biophysics researchers to upload, analyze, visualize, and share fluorescence measurement data from any location without requiring local software installation or high-performance workstations.

The existing desktop version of Full_SMS, developed by the University of Pretoria Biophysics Group, is a DearPyGui-based application that performs advanced analysis tasks including:

- Change point detection using the Watkins and Yang algorithm
- Hierarchical clustering using AHCA with BIC optimization
- Fluorescence lifetime fitting through multi-exponential decay analysis
- Auto-correlation and cross-correlation g(2) analysis
- Spectra and raster scan visualization

1.2.1 Current Challenges

The desktop application currently presents several limitations:

- Installation complexity due to Python environment configuration and dependency management
- Restricted access to machines where the software is installed
- Limited collaboration capabilities between researchers
- Inability to remotely monitor long-running analyses
- Computational limitations based on local workstation hardware

1.2.2 Proposed Solution

The web-based platform addresses these limitations through:

- Browser-based access from any device
- No local installation requirements
- Shared datasets and collaborative analysis sessions
- Persistent sessions that can be resumed later
- In-browser and email notifications for analysis completion

1.3 Project Objectives

The primary objectives of the Full_SMS Web application are:

- Improve accessibility by enabling SMS analysis through a web browser without software installation
- Support collaboration through secure dataset sharing and shared analysis workflows
- Provide an intuitive user interface with a minimal learning curve
- Improve reproducibility through detailed reports containing parameters and analysis results
- Support scalability for concurrent users and computationally intensive analysis tasks
- Ensure compatibility with HDF5 files while allowing future extensibility for additional formats

2. User Stories and User Characteristics

2.1 User Roles

The system is designed for two primary user groups.

2.1.1 Researcher (Primary User)

Researchers are academic or industry scientists conducting single-molecule spectroscopy experiments. They generate HDF5 measurement files from laboratory equipment and require tools for advanced analysis, including change point detection, clustering, lifetime fitting, and correlation analysis.

Technical Proficiency

- Moderate to advanced scientific computing knowledge
- Familiar with data analysis concepts
- Limited software development experience may be expected

Usage Characteristics

- Regular usage ranging from daily to weekly
- Ability to run multiple analyses concurrently
- Requirement for publication-quality exports
- Need for reproducible workflows and reporting

2.1.2 Collaborator (Secondary User)

Collaborators are researchers who are granted access to shared datasets and analysis sessions by other users. Their permissions may range from read-only access to limited editing capabilities.

Technical Proficiency

- Similar scientific background to primary researchers
- May be less familiar with the originating experimental setup

Usage Characteristics

- Occasional access for collaborative projects
- Primarily focused on viewing and reviewing analyses
- May export results for presentations or publications
- Restricted access to explicitly shared datasets

2.2 User Stories

The following user stories describe the system requirements from the perspective of end users.

2.2.1 Authentication and Access Control

- **US-01:** As a researcher, I want to be able to create an account and securely log into the platform so that my research data remains private and protected.
- **US-02:** As a researcher, I want to invite collaborators to access my datasets so that my team members can view and review my analysis work.
- **US-03:** As a collaborator, I want to load previously saved sessions so that I can review shared analysis work.
- **US-04:** As a collaborator, I want to request edit permissions for shared datasets so that I can contribute to collaborative analyses.

2.2.2 File Management and Data Upload

- **US-05:** As a researcher, I want to upload HDF5 measurement files so that I can begin analysis without transferring files to another workstation.
- **US-06:** As a researcher, I want support for multiple file formats so that I can work with different experimental datasets.
- **US-07:** As a researcher, I want to share datasets with collaborators without requiring additional software installation.

2.2.3 Session Management

- **US-08:** As a researcher, I want to save my analysis session so that I can continue my work later without repeating completed steps.
- **US-09:** As a researcher, I want to access the platform through any browser so that I can monitor analyses remotely.

2.2.4 Analysis and Processing

- **US-10:** As a researcher, I want analysis tasks to run in the background so that I can continue working on other datasets simultaneously.
- **US-11:** As a researcher, I want to configure default analysis parameters so that repetitive tasks require less manual setup.
- **US-12:** As a researcher, I want to receive notifications when long-running analyses are complete so that I can manage my workflow efficiently.
- **US-13:** As a researcher, I want to navigate between distinct tabs for Intensity, Lifetime, Grouping, Correlation, Spectra, Raster, and Export so that I can execute the complete standardized SMS analysis workflow.
- **US-14:** As a researcher, I want to view and interact with zoomable plots within each analysis view so that I can inspect high-precision spectroscopy data details.

2.2.5 Data Export and Reporting

- **US-15:** As a researcher, I want to export plots as high-resolution PNG images so that I can include them in presentations and publications.
- **US-16:** As a researcher, I want to export multiple files in a single operation so that I can save time when preparing analysis outputs.
- **US-17:** As a researcher, I want to generate a formatted PDF report containing analysis parameters and graphs so that I have a reproducible record of the session.
- **US-18:** As a researcher, I want to export my processed analysis results into CSV and HDF5 formats so that I can utilize them in external tools or retain native data structures.

2.2.6 Usability and User Experience

- **US-19:** As a researcher, I want the platform to be easy to learn and use so that I can focus on scientific analysis rather than software operation.

2.2.7 Data Integrity and Accountability

- **US-20:** As a researcher, I want the system to maintain a detailed log of all analysis actions performed on each dataset to guarantee reproducibility and workflow accountability.

2.2.8 Plugin Management

- **US-21:** As a plugin owner, I want to create custom analysis plugins using a code editor with parameter and output configuration so that I can extend the system's analytical capabilities for my specific research needs.
- **US-22:** As a researcher, I want to execute installed plugins on my measurement data with configurable parameters so that I can perform specialized analyses within my existing workflow.
- **US-23:** As a researcher, I want to view plugin execution results as values, plots, tables, or heatmaps so that I can interpret the analysis output in the most appropriate format.
- **US-24:** As a plugin owner, I want to submit my plugins to the marketplace for review so that other researchers can discover and use my analysis tools.
- **US-25:** As a plugin owner, I want to update my published plugins with new versions so that I can fix bugs and add improvements while maintaining compatibility for existing users.
- **US-26:** As a plugin owner, I want to delete or deprecate my plugins so that I can remove outdated or unsupported analysis tools from the marketplace.
- **US-27:** As a plugin owner, I want to monitor the approval status of my submitted plugins so that I know when my plugins are available to other users or if corrections are needed.

2.2.9 Plugin Administration

- **US-28:** As an administrator, I want to review submitted plugins before publication so that I can ensure all marketplace plugins meet security and quality standards.
- **US-29:** As an administrator, I want to approve or reject plugin submissions with feedback so that plugin owners understand the decision and can make necessary improvements.

2.2.10 Plugin Discovery and Usage

- **US-30:** As a researcher, I want to browse the plugin marketplace with filtering and search options so that I can discover analysis tools relevant to my research domain.
- **US-31:** As a researcher, I want to install plugins from the marketplace so that I can extend my analysis capabilities without writing custom code.
- **US-32:** As a researcher, I want to rate and review plugins I have used so that I can help other researchers identify high-quality analysis tools.

2.2.11 Workspace Management

- **US-33:** As a researcher, I want to create isolated workspaces with descriptive names and descriptions so that I can organize my spectroscopy experiments into logical project groupings.
- **US-34:** As a researcher, I want to view all my workspaces in a centralized dashboard, so that I can quickly access my ongoing and past research projects.
- **US-35:** As a researcher, I want to search and filter my workspaces by name, description, or status, so that I can quickly locate specific projects among many workspaces.
- **US-36:** As a researcher, I want to filter workspaces by their status (active, archived, or all), so that I can focus on current work while maintaining access to completed projects.
- **US-37:** As a researcher, I want to archive completed workspaces so that they don't clutter my active project list while remaining accessible for future reference.
- **US-38:** As a researcher, I want to unarchive previously archived workspaces, so that I can resume work on past projects when needed.
- **US-39:** As a researcher, I want to update workspace names and descriptions so that I can maintain accurate project documentation as my research evolves.
- **US-40:** As a researcher, I want to delete workspaces I no longer need, so that I can manage my storage usage and remove obsolete projects.
- **US-41:** As a researcher, I want to see the number of uploaded files in each workspace so that I can quickly assess the scope and data volume of each project.
- **US-42:** As a researcher, I want to upload HDF5 measurement files directly to a workspace so that all related experimental data is organized in one location.
- **US-43:** As a researcher, I want to view all uploaded files within a workspace so that I can select which dataset to analyze.
- **US-44:** As a researcher, I want to open uploaded files from the workspace to begin analysis so that I can transition seamlessly from file management to data processing.
- **US-45:** As a researcher, I want my workspace data to be stored securely in isolated storage paths so that my research data remains private and organized by project.

2.3 User Priorities

The following priorities have been identified for the initial implementation phase.

High Priority

- Authentication and secure login
- HDF5 file upload support
- Session save and load functionality
- Background analysis processing
- Basic plot export functionality

Medium Priority

- Support for additional file formats
- Dataset sharing and collaboration
- Batch export functionality
- PDF report generation

Lower Priority

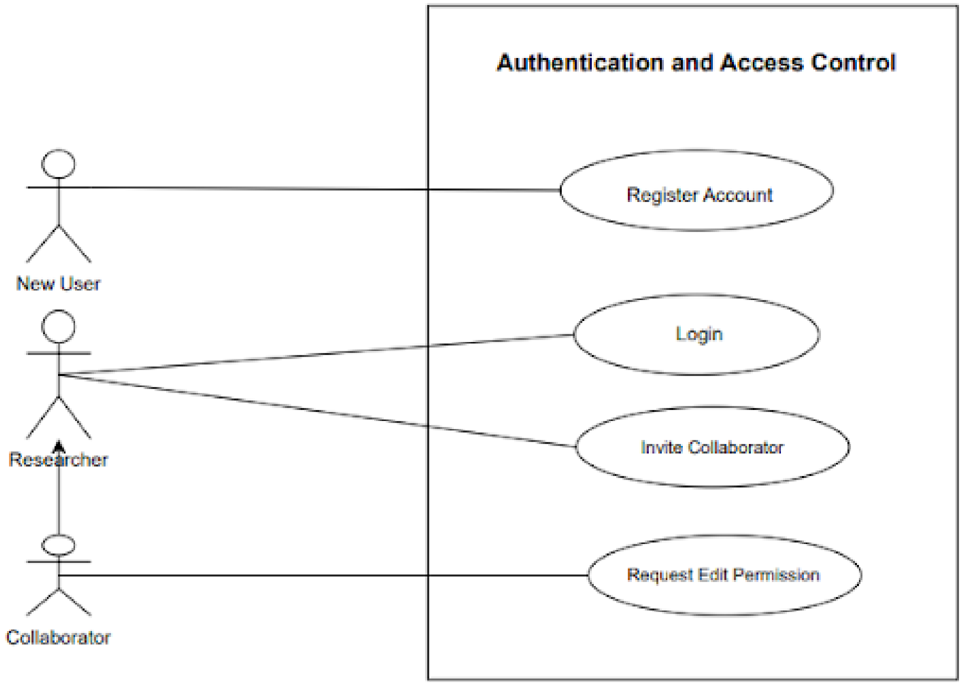
- Default parameter configuration
- Advanced collaboration features
- Email and browser notifications

3. Use Cases

This section describes the primary use cases derived from the user stories and functional requirements. Each use case defines the event that initiates the interaction and the expected outcome once the interaction is completed.

3.1 Use Case Descriptions

3.1.1 Authentication and Access Control



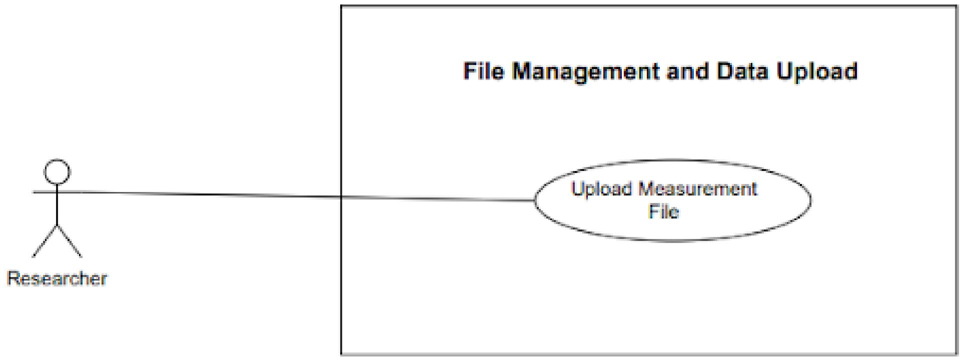
Use Case: Register Account
Related User Stories: US-01
The Use Case Begins When: A new user navigates to the Full_SMS web application and selects the "Register" option on the landing page.
The Use Case Ends When: The system successfully creates the user account and redirects the user to the login page.

Use Case: Login to System
Related User Stories: US-01, US-03
The Use Case Begins When: A registered user enters their login credentials and selects the "Login" button.
The Use Case Ends When: The user is authenticated successfully and gains access to the dashboard containing experiments that they are working on, this includes experiments they own and the ones they were invited to.

Use Case: Invite Collaborator
Related User Stories: US-02, US-07
The Use Case Begins When: A researcher selects the "Share" option for a dataset and enters a collaborator's email address.
The Use Case Ends When: The collaborator receives an invitation to access the dataset, and the researcher receives confirmation that the invitation was sent successfully.

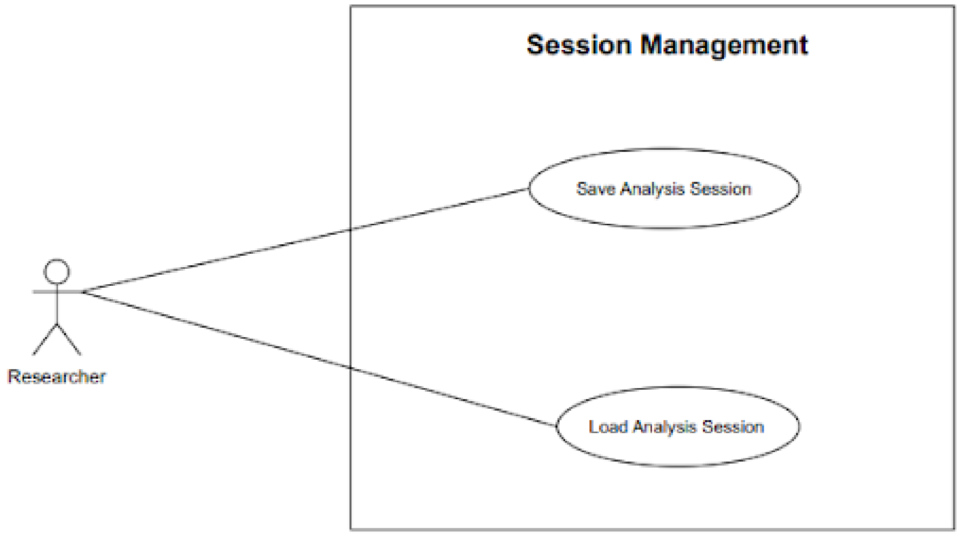
Use Case: Request Edit Permission
Related User Stories: US-04
The Use Case Begins When: A collaborator viewing a shared dataset selects the "Request Edit Access" option.
The Use Case Ends When: The dataset owner receives a notification requesting edit permissions for the collaborator.

3.1.2 File Management and Data Upload



Use Case: Upload Measurement File
Related User Stories: US-05, US-06
The Use Case Begins When: An authenticated researcher selects the "Upload File" option and chooses an HDF5 measurement file from their local machine.
The Use Case Ends When: The system uploads the file successfully, parses it according to the internal scientific data models, and displays measurement metadata, including photon counts, channel information, and acquisition parameters.

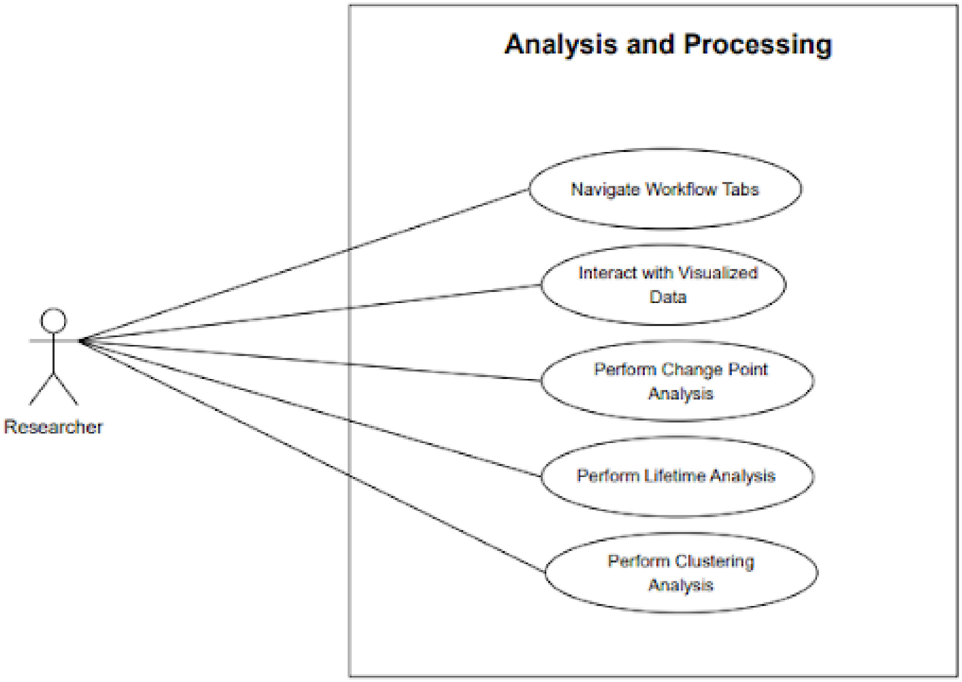
3.1.3 Session Management



Use Case: Save Analysis Session
Related User Stories: US-08
The Use Case Begins When: A researcher selects the "Save Session" option after completing one or more analyses.
The Use Case Ends When: The current analysis session is stored successfully and appears in the user's recent sessions list.

Use Case: Load Analysis Session
Related User Stories: US-03, US-08, US-09
The Use Case Begins When: A researcher or collaborator selects a saved session from the dashboard.
The Use Case Ends When: The system restores the saved session, including datasets, computed results, analysis parameters, and visualizations.

3.1.4 Analysis and Processing



Use Case: Navigate Workflow Tabs
Related User Stories: US-13
The Use Case Begins When: A researcher selects an active dataset and clicks on any of the core analysis pipeline tabs (Intensity, Lifetime, Grouping, Correlation, Spectra, Raster, or Export).
The Use Case Ends When: The application dynamically displays the selected scientific workflow view alongside its corresponding, context-specific plotting areas and parameter tools.

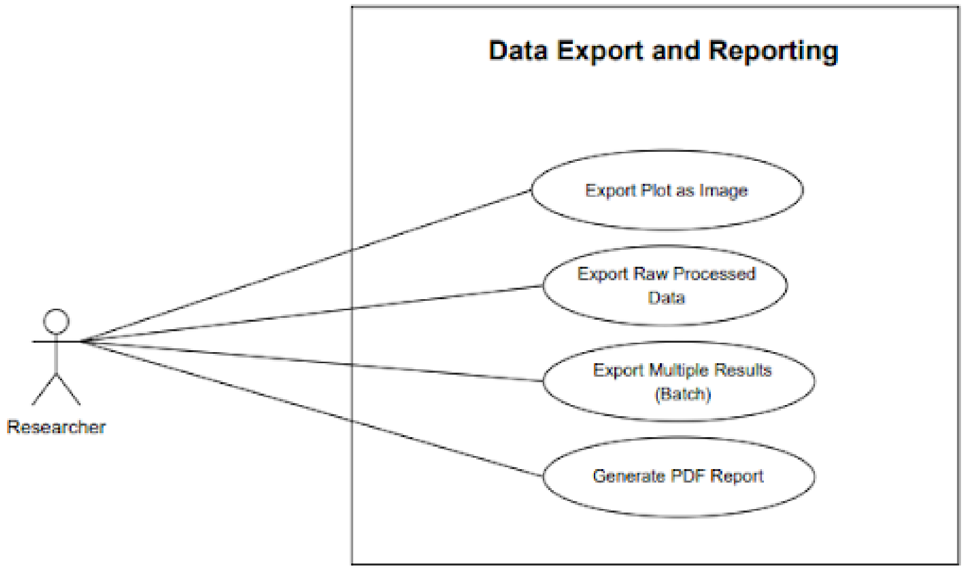
Use Case: Interact with Visualized Data
Related User Stories: US-14
The Use Case Begins When: A researcher uses mouse gestures to zoom into, pan across, or select specific regions on any generated spectroscopy plot.
The Use Case Ends When: The plot updates in high resolution instantly to isolate the selected data window, allowing high-precision inspection of data details.

Use Case: Perform Change Point Analysis
Related User Stories: US-10, US-11, US-12, US-13
The Use Case Begins When: A researcher navigates to the Intensity view, configures processing parameters, and chooses the "Run Change Point Analysis" option.
The Use Case Ends When: The system runs the algorithm asynchronously in the background, delivers a completion notification, and displays the results as an interactive intensity plot with identified photon brightness levels.

Use Case: Perform Lifetime Analysis
Related User Stories: US-10, US-11, US-12, US-13
The Use Case Begins When: A researcher navigates to the Lifetime view and selects the "Fit Lifetime Decay" option after defining the active lifetime constraints.
The Use Case Ends When: The system processes the calculation asynchronously and displays the resulting fluorescence lifetime decay histogram, fitted decay curves, and calculated fitting parameters.

Use Case: Perform Clustering Analysis
Related User Stories: US-10, US-11, US-12, US-13
The Use Case Begins When: A researcher navigates to the Grouping view and selects the "Cluster Levels" option to sort identified intensity steps.
The Use Case Ends When: The system displays the hierarchical clustering dendrogram and optimized cluster groupings calculated using Bayesian Information Criterion (BIC) analysis.

3.1.5 Data Export and Reporting



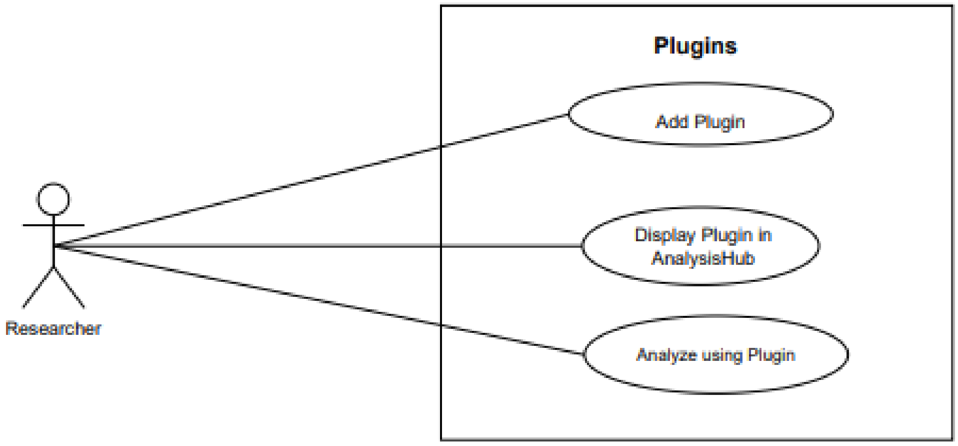
Use Case: Export Plot as Image
Related User Stories: US-15
The Use Case Begins When: A researcher selects the "Export" option while viewing an individual analysis plot and chooses PNG as the export format.
The Use Case Ends When: The system generates and downloads a high-resolution PNG image containing the current plot view and associated labels suitable for publication.

Use Case: Export Raw Processed Data
Related User Stories: US-18
The Use Case Begins When: A researcher selects the "Export" option under the dedicated Export tab and chooses CSV or HDF5 as the output data type.
The Use Case Ends When: The system compiles the processed matrices and results into the chosen file format and downloads it directly to the local storage machine.

Use Case: Export Multiple Results (Batch)
Related User Stories: US-16
The Use Case Begins When: A researcher marks multiple generated analysis assets or sessions from the Export interface and selects the "Batch Export" option.
The Use Case Ends When: The system compresses all selected exports together and downloads a structured ZIP archive containing the requested parameters and datasets.

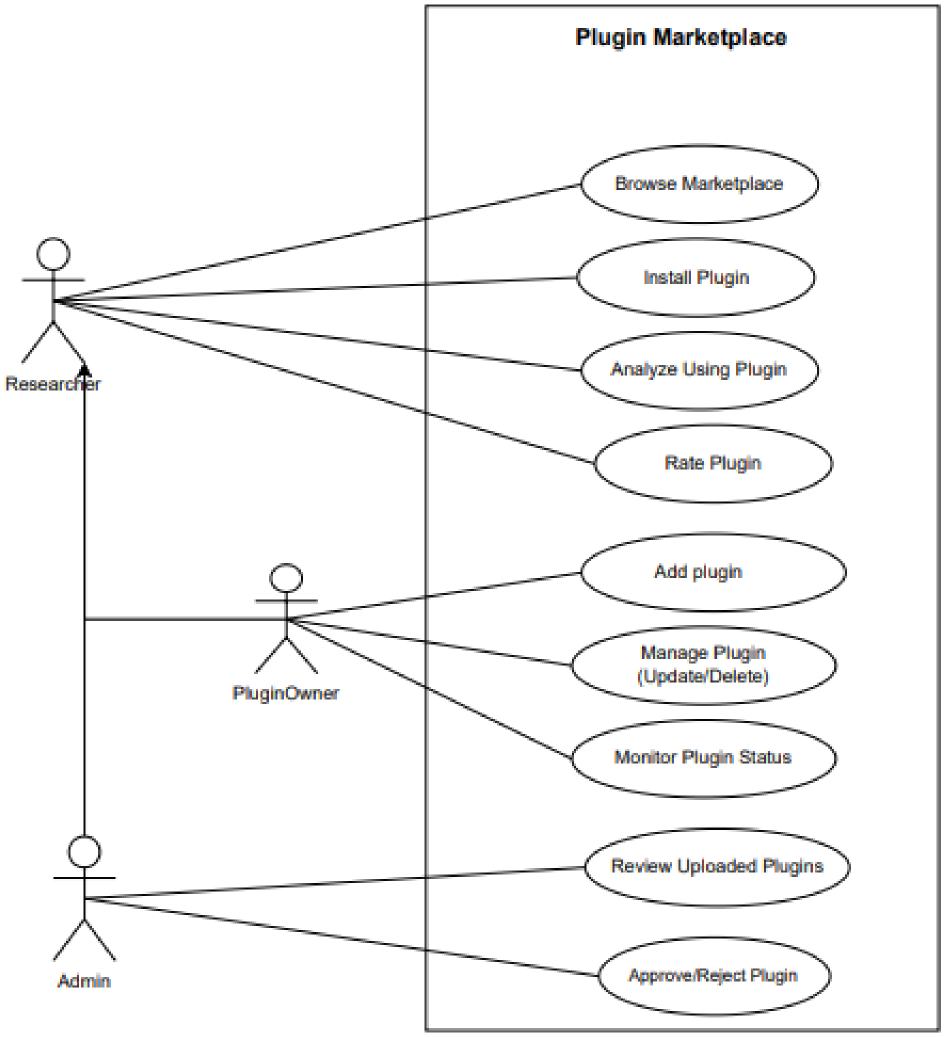
Use Case: Generate PDF Report
Related User Stories: US-17
The Use Case Begins When: A researcher selects the "Generate Report" option under the Export tab after finalizing an entire session workflow.
The Use Case Ends When: The system compiles all historical analysis configurations, input parameters, result summaries, and graphical data plots into a cleanly formatted PDF report document for academic distribution.

3.1.6 Plugin Management



Use Case: Create Plugin
Related User Stories: US-21
The Use Case Begins When: An authenticated user navigates to the Plugins and selects "Create New Plugin."
The Use Case Ends When: The system validates the plugin script for security compliance, saves the plugin configuration (parameters, outputs) and adds the plugin to the owner's plugin library.

Use Case: Execute Plugin Analysis
Related User Stories: US-22, US-23
The Use Case Begins When: A researcher selects an installed plugin from the Plugin Tab in the Analysis Hub and configures the required input parameters for a loaded measurement.
The Use Case Ends When: The system executes the plugin script in a sandboxed environment, processes the measurement data according to the plugin logic, and displays the results (values, plots, tables, or heatmaps) in the workspace analysis hub.



Use Case: Submit Plugin to Marketplace
Related User Stories: US-24, US-27
The Use Case Begins When: A plugin owner selects a completed plugin from their library and clicks "Submit for Review."
The Use Case Ends When: The system changes the plugin status to "Pending Review," notifies administrators of the new submission, and displays a confirmation to the plugin owner with the expected review timeline.

Use Case: Review Plugin Submission
Related User Stories: US-28, US-29
The Use Case Begins When: An administrator accesses the plugin review queue and selects a pending plugin submission.
The Use Case Ends When: The administrator either approves the plugin (publishing it to the marketplace) or rejects it with feedback. The system notifies the plugin owner of the decision and updates the plugin status accordingly.

Use Case: Browse Plugin Marketplace
Related User Stories: US-30
The Use Case Begins When: An authenticated researcher navigates to the Plugin Marketplace from the main dashboard or analysis hub.
The Use Case Ends When: The system displays available plugins with their descriptions, ratings, installation counts, and compatibility information, allowing the user to filter and search by category, rating, or functionality.

Use Case: Install Plugin
Related User Stories: US-31
The Use Case Begins When: A plugin user selects a plugin from the marketplace and clicks "Install."
The Use Case Ends When: The system adds the plugin to the user's installed plugins, makes it available in the Analysis Hub plugin tab, and displays a confirmation with usage instructions.

Use Case: Update Plugin
Related User Stories: US-25
The Use Case Begins When: A plugin owner selects an existing plugin from their library and clicks "Edit."
The Use Case Ends When: The system saves the updated plugin configuration and script. If the plugin was previously published, it either creates a new version pending review or updates the existing version based on the nature of changes.

Use Case: Rate Plugin
Related User Stories: US-32
The Use Case Begins When: A plugin user who has executed a plugin at least once selects the rating option from the plugin details page.
The Use Case Ends When: The system records the user's rating, updates the plugin's average rating, and optionally saves any written review provided by the user.

Use Case: Delete Plugin
Related User Stories: US-26
The Use Case Begins When: A plugin owner selects an owned plugin and clicks Delete.
The Use Case Ends When: The system prompts for confirmation, removes the plugin from the marketplace (if published), notifies users who have installed the plugin of its deprecation, and archives the plugin data.

3.1.7 Workspace Management

Use Case: Create Workspace
Related User Stories: US-33, US-45
The Use Case Begins When: An authenticated researcher navigates to the dashboard and selects the "New Workspace" button.
The Use Case Ends When: The system creates a new workspace with a unique identifier, assigns an isolated storage bucket path, sets the status to "active," and displays the workspace in the researcher's dashboard with zero file count.

Use Case: View All Workspaces
Related User Stories: US-34, US-41
The Use Case Begins When: An authenticated researcher navigates to the dashboard after logging in.
The Use Case Ends When: The system displays a table or grid view of all workspaces owned by the researcher, showing workspace name, description, status, creation date, last updated date, and file count for each workspace.

Use Case: Search Workspaces
Related User Stories: US-35
The Use Case Begins When: A researcher enters text into the workspace search field on the dashboard.
The Use Case Ends When: The system filters the displayed workspaces in real-time, showing only those with names or descriptions matching the search query.

Use Case: Filter Workspaces by Status
Related User Stories: US-36
The Use Case Begins When: A researcher selects a status filter button (Active, Archived, or All) on the dashboard.
The Use Case Ends When: The system updates the workspace list to display only workspaces matching the selected status, showing the count of workspaces in each category.

Use Case: View Workspace Details
Related User Stories: US-34, US-41, US-43
The Use Case Begins When: A researcher selects a specific workspace from the dashboard by clicking on it.
The Use Case Ends When: The system navigates to the workspace detail page, displaying the workspace name, description, status badge, and a list of all uploaded HDF5 files with their filenames and sizes.

Use Case: Update Workspace Information
Related User Stories: US-39
The Use Case Begins When: A researcher selects the "Edit" option for a workspace and modifies the workspace name or description.
The Use Case Ends When: The system validates the new information (minimum 3 characters for name), updates the workspace record, sets a new updated_at timestamp, and displays a confirmation message.

Use Case: Archive Workspace
Related User Stories: US-37
The Use Case Begins When: A researcher selects the "Archive" option for an active workspace from the dashboard or workspace menu.
The Use Case Ends When: The system changes the workspace status from "active" to "archived," updates the updated_at timestamp, and moves the workspace to the archived filter view while maintaining all file associations.

Use Case: Unarchive Workspace
Related User Stories: US-38
The Use Case Begins When: A researcher viewing archived workspaces selects the "Unarchive" option for a specific workspace.
The Use Case Ends When: The system changes the workspace status from "archived" to "active," updates the updated_at timestamp, and returns the workspace to the active workspace list.

Use Case: Delete Workspace
Related User Stories: US-40
The Use Case Begins When: A researcher selects the "Delete" option for a workspace and confirms the deletion in the confirmation dialog.
The Use Case Ends When: The system permanently removes the workspace record from the database, deletes all associated files from the storage bucket at the workspace's storage_bucket_path, and removes the workspace from the dashboard display.

Use Case: Upload File to Workspace
Related User Stories: US-42, US-05, US-06
The Use Case Begins When: A researcher viewing a workspace selects the "Upload File" button and chooses an HDF5 file from their local machine.
The Use Case Ends When: The system uploads the file to the workspace's dedicated storage bucket path, creates an upload record linked to the workspace_id, increments the workspace file count, and displays the newly uploaded file in the workspace file list with its filename and size.

Use Case: View Workspace Uploads
Related User Stories: US-43, US-41
The Use Case Begins When: A researcher navigates to a workspace detail page.
The Use Case Ends When: The system queries all HDF5 upload records associated with the workspace_id and displays them as interactive cards showing filename and file size in megabytes, or shows an empty state message if no files exist.

Use Case: Open Upload for Analysis
Related User Stories: US-44, US-13
The Use Case Begins When: A researcher clicks on an uploaded file card within a workspace.
The Use Case Ends When: The system sets the selected upload as the current active upload, navigates to the Analysis Hub, and loads the measurement data for scientific workflow analysis across all pipeline tabs.

4. Functional Requirements

4.1 R1: User Management and Security

4.1.1 R1.1: User Authentication

- **R1.1.1:** The system shall allow users to create accounts using a valid email address and password.
- **R1.1.2:** The system shall enforce password strength requirements (minimum length, uppercase, numeric, and special characters).
- **R1.1.3:** The system shall validate email address format during registration.
- **R1.1.4:** The system shall authenticate users using email and password credentials.
- **R1.1.5:** The system shall issue an authenticated session upon successful login.
- **R1.1.6:** The system shall maintain the authenticated session until the user logs out or the session expires.
- **R1.1.7:** The system shall provide a secure password reset mechanism using time-limited reset links.
- **R1.1.8:** The system shall invalidate password reset links after use or expiration.

4.1.2 R1.2: Role-Based Access Control (RBAC)

- **R1.2.1:** The system shall support at least two user roles: Administrator and Researcher.
- **R1.2.2:** The system shall restrict protected functionality to authenticated users only.
- **R1.2.3:** The system shall enforce role-based permissions for datasets, analysis tools, and administrative features.
- **R1.2.4:** The system shall deny unauthorized access attempts.
- **R1.2.5:** Administrators shall be able to assign roles to users.
- **R1.2.6:** Administrators shall be able to modify roles assigned to users.
- **R1.2.7:** Administrators shall be able to revoke roles from users.

4.1.3 R1.3: Session Management

- **R1.3.1:** The system shall allow authenticated users to save the current analysis session.
- **R1.3.2:** Saved sessions shall include dataset references, analysis parameters, regions of interest, computed results, and visualizations.
- **R1.3.3:** The system shall display saved sessions in the user's list of recent sessions.
- **R1.3.4:** Researchers and authorized collaborators shall be able to load a saved session from the dashboard.
- **R1.3.5:** The system shall restore all associated datasets, computed results, analysis parameters, and visualizations upon loading a saved session.
- **R1.3.6:** Users shall be able to resume saved sessions from any supported browser.

4.1.4 R1.4: Default Analysis Parameters

- **R1.4.1:** The system shall allow users to save reusable analysis parameter configurations (e.g., binning width, clustering bounds).

- **R1.4.2:** Users shall be able to load saved configurations before analysis execution.

4.1.5 R1.5: Dataset Sharing

- **R1.5.1:** The system shall allow dataset owners to share datasets with other registered users.
- **R1.5.2:** Dataset sharing shall support multiple access levels (read-only, edit).
- **R1.5.3:** Dataset owners shall be able to revoke shared access at any time.

4.2 R2: Data Management and Ingestion

4.2.1 R2.1: File Upload

- **R2.1.1:** Authenticated users shall be able to upload spectroscopy data in .pt3 and .csv formats.
- **R2.1.2:** The system shall automatically intercept and reject unsupported file type extensions.
- **R2.1.3:** The system shall track and display upload progress for each file added to the queue.

4.2.2 R2.2: Automatic Data Conversion

- **R2.2.1:** The system shall automatically convert uploaded datasets into internal HDF5 (.h5) format.
- **R2.2.2:** The system shall notify users upon successful conversion completion.
- **R2.2.3:** The system shall notify users upon conversion failure.

4.2.3 R2.3: Persistent Storage

- **R2.3.1:** The system shall store converted datasets and metadata (size, owner, timestamp) in centralized storage.

4.2.4 R2.4: Data Export

- **R2.4.1:** Users shall be able to export results in CSV format.
- **R2.4.2:** Users shall be able to export results in PNG format.
- **R2.4.3:** Users shall be able to export results in Parquet format.
- **R2.4.4:** Users shall be able to export results in PDF format.
- **R2.4.5:** Users shall be able to export results in SVG format.
- **R2.4.6:** Users shall be able to export results in Excel format.
- **R2.4.7:** Users shall be able to export results in JSON format.

4.2.5 R2.5: Bulk and Batch Export

- **R2.5.1:** The system shall support simultaneous multi-dataset exports packaged as ZIP archives.

4.2.6 R2.6: Temporary File Management

- **R2.6.1:** The system shall automatically purge temporary conversion files after a retention period.

4.3 R3: Signal Analysis and User Controls

4.3.1 R3.1: Artifact Filtering

- **R3.1.1:** The system shall provide automated artifact detection (spikes/bursts).
- **R3.1.2:** The system shall allow users to manually mark artifacts.

4.3.2 R3.2: Automated State Detection

- **R3.2.1:** The system shall perform Change-Point Analysis (CPA) on photon traces.
- **R3.2.2:** The system shall cluster detected states from the CPA output.

4.3.3 R3.3: Parameter Adjustment

- **R3.3.1:** Users shall adjust binning width via input controls.
- **R3.3.2:** Users shall adjust clustering bounds via input controls.

4.3.4 R3.4: Background Processing

- **R3.4.1:** Intensive analyses (clustering, fitting) shall execute asynchronously via task queues.

4.4 R4: Visualization and Reporting

4.4.1 R4.1: Interactive Time-Trace Display

- **R4.1.1:** The system shall render zoomable time-trace plots for single and dual-channel datasets.

4.4.2 R4.2: Raster Scan Image Rendering

- **R4.2.1:** The system shall generate 2D raster scans with interactive tooltips for coordinates and intensity.

4.4.3 R4.3: Statistical Chart Generation

- **R4.3.1:** The system shall generate intensity histograms.
- **R4.3.2:** The system shall generate BIC plots.

4.4.4 R4.4: Notification System

- **R4.4.1:** The system shall provide in-browser notifications for task status.
- **R4.4.2:** The system shall provide optional email notifications for task status.

4.5 R5: Administrative and Monitoring Features

4.5.1 R5.1: Job and Queue Monitoring

- **R5.1.1:** Administrators shall have an interface to monitor active, pending, or completed jobs.

4.5.2 R5.2: Audit and Activity Logging

- **R5.2.1:** The system shall maintain immutable logs of all dataset actions (uploads, edits, sharing).

4.6 R6: Workspace Management

4.6.1 R6.1: Workspace Creation

- **R6.1.1:** The system shall allow authenticated researchers to create a workspace with a name and description.
- **R6.1.2:** The system shall set a newly created workspace's status to "active" by default.
- **R6.1.3:** The system shall display all workspaces owned by a researcher in a dashboard view.

4.6.2 R6.2: Workspace Editing

- **R6.2.1:** Researchers shall be able to update a workspace's name and description.
- **R6.2.2:** The system shall validate updated workspace names against a minimum length requirement.

4.6.3 R6.3: Workspace Archiving

- **R6.3.1:** Researchers shall be able to archive an active workspace.
- **R6.3.2:** Researchers shall be able to unarchive an archived workspace.

4.6.4 R6.4: Workspace Deletion

- **R6.4.1:** Researchers shall be able to permanently delete a workspace.

4.6.5 R6.5: Workspace File Management

- **R6.5.1:** Researchers shall be able to upload HDF5 measurement files to a workspace.
- **R6.5.2:** Researchers shall be able to view all uploaded files within a workspace.

4.7 R7: Plugin Management

4.7.1 R7.1: Plugin Creation

- **R7.1.1:** Plugin owners shall be able to create custom analysis plugins using a code editor.
- **R7.1.2:** Plugin owners shall be able to configure input parameters for a plugin.
- **R7.1.3:** Plugin owners shall be able to configure the output format for a plugin.

4.7.2 R7.2: Plugin Execution

- **R7.2.1:** Researchers shall be able to execute installed plugins on their measurement data.
- **R7.2.2:** Researchers shall be able to configure parameters before executing a plugin.

5. Quality (Non-Functional) Requirements

Quality requirements (non-functional requirements) define system characteristics and constraints that are not directly related to functional behavior. These requirements ensure the system is reliable, performant, secure, usable and maintainable.

5.1 NFR1: Performance Requirements

5.1.1 NFR1.1: Page Load Performance

- **NFR1.1.1:** The system shall load pages with a Largest Contentful Paint (LCP) of 2.1 seconds or less, and a First Contentful Paint (FCP) of under 1 second, measured via Lighthouse under standard network conditions.
- **NFR1.1.2:** The system shall maintain a Total Blocking Time of under 100ms and a Cumulative Layout Shift (CLS) of 0, ensuring no visual instability during load.

5.1.2 NFR1.2: File Upload Capability

- **NFR1.2.1:** The system shall support uploads of large files with real-time progress tracking.

5.1.3 NFR1.3: Concurrent User Capacity

- **NFR1.3.1:** The system shall maintain consistent performance across multiple concurrent user sessions without noticeable degradation.

5.1.4 NFR1.4: Background Processing Efficiency

- **NFR1.4.1:** The system shall complete background analysis tasks within a reasonable timeframe proportional to algorithm complexity.

5.2 NFR2: Reliability Requirements

5.2.1 NFR2.1: System Availability

- **NFR2.1.1:** The system shall target 99% uptime during evaluation.

5.2.2 NFR2.2: Task Recovery

- **NFR2.2.1:** The system shall automatically recover from failed asynchronous tasks within acceptable timeframes.

5.2.3 NFR2.3: Data Backup and Retention

- **NFR2.3.1:** The system shall perform regular automated backups of all data.
- **NFR2.3.2:** The system shall retain backup copies for an extended retention period to enable recovery.

5.2.4 NFR2.4: Session Persistence

- **NFR2.4.1:** The system shall preserve user session data across infrastructure restarts.

5.3 NFR3: Scalability Requirements

5.3.1 NFR3.1: Large Dataset Support

- **NFR3.1.1:** The system shall handle analysis of large-scale datasets.

5.3.2 NFR3.2: User Data Capacity

- **NFR3.2.1:** The system shall store and manage a substantial number of datasets per user without performance degradation.

5.3.3 NFR3.3: Horizontal Scaling

- **NFR3.3.1:** The system shall support horizontal scaling through containerization and load distribution.

5.3.4 NFR3.4: Storage Capacity

- **NFR3.4.1:** The system shall scale storage infrastructure to accommodate organizational growth.

5.4 NFR4: Security Requirements

5.4.1 NFR4.1: Data Encryption in Transit

- **NFR4.1.1:** The system shall encrypt all network communications using industry-standard encryption protocols.

5.4.2 NFR4.2: Authentication Token Management

- **NFR4.2.1:** The system shall issue authentication tokens with defined expiration periods.
- **NFR4.2.2:** The system shall provide mechanisms for token refresh without re-authentication.

5.4.3 NFR4.3: Password Security

- **NFR4.3.1:** The system shall store passwords using industry-standard cryptographic hashing algorithms such as ES256.

5.4.4 NFR4.4: SQL Injection Prevention

- **NFR4.4.1:** The system shall prevent SQL injection attacks through parameterized query execution.

5.4.5 NFR4.5: Cross-Site Scripting (XSS) Protection

- **NFR4.5.1:** The system shall implement security headers to prevent cross-site scripting attacks.

5.4.6 NFR4.6: Cross-Origin Resource Sharing (CORS)

- **NFR4.6.1:** The system shall restrict cross-origin requests to approved sources only.

5.4.7 NFR4.7: File Upload Validation

- **NFR4.7.1:** The system shall validate uploaded files against a whitelist of permitted file types.

5.5 NFR5: Maintainability Requirements

5.5.1 NFR5.1: Code Coverage

- **NFR5.1.1:** The system shall maintain a minimum of 50% overall automated backend test coverage, with newly developed service and controller components maintaining higher coverage individually (target $\geq 70\%$) as legacy scientific analysis modules are incrementally covered in upcoming iterations.

5.5.2 NFR5.2: API Documentation

- **NFR5.2.1:** The system shall provide comprehensive API documentation in a standardized format.

5.5.3 NFR5.3: Code Documentation

- **NFR5.3.1:** The system shall include documentation for all functions and classes via inline comments.

5.5.4 NFR5.4: Continuous Integration and Deployment

- **NFR5.4.1:** The system shall validate builds automatically before deployment to production.

5.5.5 NFR5.5: Error Logging

- **NFR5.5.1:** The system shall capture detailed error information including stack traces and contextual data for debugging.

5.6 NFR6: Usability Requirements

5.6.1 NFR6.1: Browser Compatibility

- **NFR6.1.1:** The system shall function consistently across modern versions of major web browsers.

5.6.2 NFR6.2: User-Friendly Error Messages

- **NFR6.2.1:** The system shall display error messages using clear, non-technical language.

5.6.3 NFR6.3: Loading State Feedback

- **NFR6.3.1:** The system shall provide visual feedback for operations that require noticeable processing time.

6. Domain Model

The domain model represents the core entities and their relationships within the Full SMS Web Application. This model captures the essential concepts, attributes, and associations that define the system's data structure and business logic.